

SQL Homework Exercises

Days 1 and 2 · MySQL 8.0 · Sales order dataset

Twelve short exercises on the sales database from Days 1 and 2. They start easy and build up. Try each one on your own before looking at anything else.

How to use this sheet

- Write your query in Workbench and run it. Every question has a **Check yourself** line, so you can tell whether your answer is right without seeing the solution.
- Stuck? Click **Show hint**. A hint names the clauses you need; it does not give the answer.
- Click **Show solution** to see the code and its result. Each solution is on its own page, and **Back to question** returns you to where you were.

Before you start

Select the database and confirm the tables are loaded. If your counts differ, run `sql_day2_setup.sql` first.

```
USE sales_db;

SELECT 'customers' AS table_name, COUNT(*) AS row_count FROM customers
UNION ALL
SELECT 'products', COUNT(*) FROM products
UNION ALL
SELECT 'orders', COUNT(*) FROM orders
UNION ALL
SELECT 'order_items', COUNT(*) FROM order_items;
```

RESULT

table_name	row_count
customers	730
products	1615
orders	1980
order_items	4000

Section A: Reading data

Question 1 WHERE

List the name and segment of every customer in the Corporate segment.

Check yourself: 212 rows.

[Show hint »](#) [Show solution »](#)

Question 2 DISTINCT, ORDER BY

Which sub-categories belong to the Furniture category? List each one once, in alphabetical order.

Check yourself: 4 rows.

[Show hint »](#) [Show solution »](#)

Question 3 COUNT

How many orders were shipped to California?

Check yourself: one row holding one number, 382.

[Show hint »](#) [Show solution »](#)

Section B: Sorting, filtering and summarising**Question 4** AND, ORDER BY, LIMIT

Show the order ID, order date and state of the five earliest orders that were shipped Second Class to the West region.

Check yourself: 5 rows. The first is order CA-2014-111059, placed on 2014-02-03.

[Show hint »](#) [Show solution »](#)

Question 5 GROUP BY

How many orders were placed in each year? Show the year and the number of orders, earliest year first.

Check yourself: 4 rows. 2017 has 658 orders.

[Show hint »](#) [Show solution »](#)

Question 6 HAVING

Which cities have received more than 50 orders? Show the city and its number of orders, busiest first.

Check yourself: 8 rows. New York City is first with 180 orders.

[Show hint »](#) [Show solution »](#)

Question 7 CASE

Classify every order by how quickly it was shipped: Fast if it shipped within 2 days of the order date, Normal if within 5 days, and Slow otherwise. How many orders fall in each class?

Check yourself: 3 rows. Normal is the largest class with 1163 orders.

[Show hint »](#) [Show solution »](#)

Section C: Combining tables and changing data**Question 8** JOIN

Show the product name and the sales value of the five largest order lines.

Check yourself: 5 rows. The first is a Cisco videoconferencing unit with sales of 22638.4800.

[Show hint »](#) [Show solution »](#)

Question 9 JOIN, GROUP BY

What are the total sales and the total profit for each customer segment? Show the segment with the highest sales first.

Check yourself: 3 rows. Consumer is first with total sales of 450350.6588.

[Show hint »](#) [Show solution »](#)

Question 10 Subquery

Which customers placed an order on the last day in the data? Show the customer's name, the order ID and the order date. Do not type the date yourself: let a subquery find it.

Check yourself: 3 rows, all dated 2017-12-30.

[Show hint »](#) [Show solution »](#)

Question 11 INSERT, UPDATE, DELETE

Practise changing data without disturbing the dataset. (a) Add a customer with the ID ZZ-00002, your own name and the segment Consumer. (b) Change that customer's segment to Home Office. (c) Check the change with a SELECT. (d) Delete the customer.

Check yourself: after step (d), `SELECT COUNT(*) FROM customers;` returns 730 again.

[Show hint »](#) [Show solution »](#)

Question 12 Stored procedure

Write a stored procedure called `top_products` that takes a category name and returns the five products with the highest total sales in that category, showing the product name and the total. Call it for Furniture.

Check yourself: 5 rows. The first is Global Commerce Series High-Back Swivel/Tilt Chairs with 6184.0660.

[Show hint »](#) [Show solution »](#)

Hints

A hint names the tools for the job. Go back and try again before opening the solution.

Hint 1: You only need the `customers` table. Filter with `WHERE` on the `segment` column. Text values go in single quotes.

[« Back to question 1](#) [Show solution »](#)

Hint 2: Use the `products` table. `DISTINCT` removes the repeats, `WHERE` picks the category and `ORDER BY` sorts the result.

[« Back to question 2](#) [Show solution »](#)

Hint 3: `COUNT(*)` counts rows. Filter the `orders` table on `state` so that only the California rows are counted.

[« Back to question 3](#) [Show solution »](#)

Hint 4: Two conditions joined with `AND`. Sort by `order_date` (ascending puts the earliest first), then keep five rows with `LIMIT`.

[« Back to question 4](#) [Show solution »](#)

Hint 5: `YEAR(order_date)` returns the year of a date. Give it an alias, group by it, and count the rows in each group with `COUNT(*)`.

[« Back to question 5](#) [Show solution »](#)

Hint 6: Group by `city` and count. A test on a count cannot go in `WHERE`; it goes in `HAVING`, after the `GROUP BY`.

[« Back to question 6](#) [Show solution »](#)

Hint 7: `DATEDIFF(ship_date, order_date)` gives the number of days. Put it in a `CASE` with two `WHEN` lines and an `ELSE`, give the `CASE` an alias, then group by that alias.

[« Back to question 7](#) [Show solution »](#)

Hint 8: Sales are in `order_items` and names are in `products`. Join the two on `product_id`, sort by sales descending and keep five rows.

[« Back to question 8](#) [Show solution »](#)

Hint 9: The `segment` is in `customers` and the money is in `order_items`, with `orders` in between. Join all three (`customer_id`, then `order_id`), group by `segment` and use `SUM`.

[« Back to question 9](#) [Show solution »](#)

Hint 10: `SELECT MAX(order_date) FROM orders` finds the last day. Put it in brackets and compare `order_date` with it in the `WHERE`. Join `customers` to get the name.

[« Back to question 10](#) [Show solution »](#)

Hint 11: Three statements: `INSERT INTO ... VALUES`, `UPDATE ... SET ... WHERE` and `DELETE FROM ... WHERE`. Use `customer_id = 'ZZ-00002'` in every `WHERE` so that only your row is touched.

« [Back to question 11](#) [Show solution »](#)

Hint 12: First write it as an ordinary query for 'Furniture': join `order_items` to `products`, filter on category, group by product, `SUM(sales)`, sort and limit. Once that works, wrap it in `CREATE PROCEDURE` and replace 'Furniture' with a parameter.

« [Back to question 12](#) [Show solution »](#)

Solution 1

Question: List the name and segment of every customer in the **Corporate** segment.

```
SELECT customer_name, segment
FROM customers
WHERE segment = 'Corporate';
```

RESULT

customer_name	segment
Ann Blume	Corporate
Alyssa Crouse	Corporate
Ann Chong	Corporate

First 3 of 212 rows shown.

WHERE keeps only the rows whose segment is Corporate.

[« Back to question 1](#) [See hint 1](#)

Solution 2

Question: Which sub-categories belong to the Furniture category? List each one once, in alphabetical order.

```
SELECT DISTINCT sub_category
FROM products
WHERE category = 'Furniture'
ORDER BY sub_category;
```

RESULT

sub_category

Bookcases

Chairs

Furnishings

Tables

Without DISTINCT each sub-category would appear once for every product in it.

[« Back to question 2](#) [See hint 2](#)

Solution 3

Question: How many orders were shipped to California?

```
SELECT COUNT(*) AS california_orders
FROM orders
WHERE state = 'California';
```

RESULT

california_orders

382

The WHERE runs first, so COUNT (*) only counts the rows that are left.

[« Back to question 3](#) [See hint 3](#)

Solution 4

Question: Show the order ID, order date and state of the five earliest orders that were shipped **Second Class** to the **West** region.

```
SELECT order_id, order_date, state
FROM orders
WHERE region = 'West'
      AND ship_mode = 'Second Class'
ORDER BY order_date
LIMIT 5;
```

RESULT

order_id	order_date	state
CA-2014-111059	2014-02-03	Washington
CA-2014-104269	2014-03-01	Washington
CA-2014-120838	2014-03-23	California
US-2014-157021	2014-04-01	California
CA-2014-167164	2014-05-13	Utah

Both conditions must be true, so they are joined with **AND**. Ascending is the default sort order, which puts the earliest date first.

[« Back to question 4](#) [See hint 4](#)

Solution 5

Question: How many orders were placed in each year? Show the year and the number of orders, earliest year first.

```
SELECT YEAR(order_date) AS order_year,  
       COUNT(*)        AS orders  
FROM orders  
GROUP BY order_year  
ORDER BY order_year;
```

RESULT

order_year	orders
2014	386
2015	398
2016	538
2017	658

Every order in the same year lands in the same group, and `COUNT (*)` counts each group.

[« Back to question 5](#) [See hint 5](#)

Solution 6

Question: Which cities have received more than 50 orders? Show the city and its number of orders, busiest first.

```
SELECT city, COUNT(*) AS orders
FROM orders
GROUP BY city
HAVING COUNT(*) > 50
ORDER BY orders DESC;
```

RESULT

city	orders
New York City	180
Los Angeles	136
Philadelphia	111
San Francisco	101
Seattle	78
Houston	69
Chicago	60
Columbus	51

HAVING filters the groups after they have been counted. Putting `COUNT(*) > 50` in a WHERE gives Error Code: 1111. Invalid use of group function.

[« Back to question 6](#) [See hint 6](#)

Solution 7

Question: Classify every order by how quickly it was shipped: **Fast** if it shipped within 2 days of the order date, **Normal** if within 5 days, and **Slow** otherwise. How many orders fall in each class?

```
SELECT CASE
    WHEN DATEDIFF(ship_date, order_date) <= 2 THEN 'Fast'
    WHEN DATEDIFF(ship_date, order_date) <= 5 THEN 'Normal'
    ELSE 'Slow'
END AS shipping_speed,
COUNT(*) AS orders
FROM orders
GROUP BY shipping_speed
ORDER BY orders DESC;
```

RESULT

shipping_speed	orders
Normal	1163
Fast	451
Slow	366

CASE stops at the first WHEN that is true, so an order shipped in 1 day is **Fast** and never reaches the second line.

« [Back to question 7](#) [See hint 7](#)

Solution 8

Question: Show the product name and the sales value of the five largest order lines.

```
SELECT p.product_name, oi.sales
FROM order_items AS oi
JOIN products AS p ON p.product_id = oi.product_id
ORDER BY oi.sales DESC
LIMIT 5;
```

RESULT

product_name	sales
Cisco TelePresence System EX90 Videoconferencing Unit	22638.4800
Canon imageCLASS 2200 Advanced Copier	11199.9680
High Speed Automatic Electric Letter Opener	8187.6500
Lexmark MX611dhe Monochrome Laser Printer	8159.9520
Cubify CubeX 3D Printer Triple Head Print	7999.9800

The join follows the foreign key `order_items.product_id` to fetch each line's product name.

[« Back to question 8](#) [See hint 8](#)

Solution 9

Question: What are the total sales and the total profit for each customer segment? Show the segment with the highest sales first.

```
SELECT c.segment,  
       SUM(oi.sales) AS total_sales,  
       SUM(oi.profit) AS total_profit  
FROM customers AS c  
JOIN orders      AS o  ON o.customer_id = c.customer_id  
JOIN order_items AS oi ON oi.order_id   = o.order_id  
GROUP BY c.segment  
ORDER BY total_sales DESC;
```

RESULT

segment	total_sales	total_profit
Consumer	450350.6588	48034.3560
Corporate	272042.7097	26952.8737
Home Office	200909.7845	21124.8540

customers and order_items are not linked directly, so the query passes through orders to connect them.

[« Back to question 9](#) [See hint 9](#)

Solution 10

Question: Which customers placed an order on the last day in the data? Show the customer's name, the order ID and the order date. Do not type the date yourself: let a subquery find it.

```
SELECT c.customer_name, o.order_id, o.order_date
FROM orders AS o
JOIN customers AS c ON c.customer_id = o.customer_id
WHERE o.order_date = (SELECT MAX(order_date) FROM orders)
ORDER BY c.customer_name;
```

RESULT

customer_name	order_id	order_date
Chuck Clark	CA-2017-126221	2017-12-30
Erica Bern	CA-2017-115427	2017-12-30
Patrick O'Donnell	CA-2017-143259	2017-12-30

The subquery returns one date. If new orders are loaded next month, the same query still finds the last day without being edited.

[« Back to question 10](#)

[See hint 10](#)

Solution 11

Question: Practise changing data without disturbing the dataset. (a) Add a customer with the ID ZZ-00002, your own name and the segment **Consumer**. (b) Change that customer's segment to **Home Office**. (c) Check the change with a **SELECT**. (d) Delete the customer.

```
-- (a)
INSERT INTO customers (customer_id, customer_name, segment)
VALUES ('ZZ-00002', 'Your Name', 'Consumer');

-- (b)
UPDATE customers
SET segment = 'Home Office'
WHERE customer_id = 'ZZ-00002';

-- (c)
SELECT * FROM customers
WHERE customer_id = 'ZZ-00002';

-- (d)
DELETE FROM customers
WHERE customer_id = 'ZZ-00002';
```

RESULT

customer_id	customer_name	segment
ZZ-00002	Your Name	Home Office

The result shown is from step (c). Every **WHERE** uses the primary key, so exactly one row is changed and then removed, and the table is back to 730 customers.

« [Back to question 11](#) [See hint 11](#)

Solution 12

Question: Write a stored procedure called `top_products` that takes a category name and returns the five products with the highest total sales in that category, showing the product name and the total. Call it for Furniture.

```
DROP PROCEDURE IF EXISTS top_products;

DELIMITER //
CREATE PROCEDURE top_products (IN p_category VARCHAR(30))
BEGIN
    SELECT p.product_name,
           SUM(oi.sales) AS total_sales
    FROM order_items AS oi
    JOIN products AS p ON p.product_id = oi.product_id
    WHERE p.category = p_category
    GROUP BY p.product_id, p.product_name
    ORDER BY total_sales DESC
    LIMIT 5;
END //
DELIMITER ;

CALL top_products('Furniture');
```

RESULT

product_name	total_sales
Global Commerce Series High-Back Swivel/Tilt Chairs	6184.0660
Bretford Rectangular Conference Table Tops	6112.1125
Hon 4070 Series Pagoda Armless Upholstered Stacking Chairs	5542.8700
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	5479.6956
SAFCO Arco Folding Chair	5413.5200

In Workbench, select the whole block from `DELIMITER //` to `DELIMITER ;` and run it with Ctrl+Shift+Enter. Then try `CALL top_products('Technology');`.

[« Back to question 12](#)

[See hint 12](#)